Kyle Fairbanks

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Mechanical Engineering graduate with experience in structural design, composite analysis, prototyping, and electro-mechanical hardware development. Led wing design, analysis, and manufacturing for a modular UAV capstone and delivered production-facing test systems, mounting hardware, and automation projects during an internship at ASSA ABLOY.

Education

University of Arizona – Tucson, AZ

- B.S. Mechanical Engineering; Minor: Aerospace Engineering — Graduated May 2026

Experience

ASSA ABLOY – Quality Engineering Intern (June 2024 - Present)

Operator Guidance and Verification System

- Designed a multi-station operator guidance and verification system using LabJack I/O, lasers, LEDs, and switches for real-time assembly validation
- Developed a self-contained Python/Flask control interface with configurable test modes, logging, error handling, and hardware-in-the-loop validation

Control ID Hub Mounting Bracket

- Designed a 3D-printable press-fit mounting bracket in SolidWorks for a production field-installation need, balancing fit, wire routing, removability, printability, and UL compliance constraints
- Advanced the bracket through prototype iterations and review to a release-ready part approved for inclusion in the product kit

Multi-Door Automated Cycle Tester

- Delivered an automated cycling tester enabling 12 parallel door-cycle tests in the footprint of one existing rig
- Integrated hydraulic, electronic, and mechanical subsystems in a modular architecture supporting automated cycling, monitoring, maintenance, and future expansion

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Projects

Dynamically Scaled Modular UAV — Senior Design Capstone, University of Arizona

- Designed a composite wing for a 1.38 m span UAV targeting 80 g mass under a 5G load requirement
- Built a custom Python structural analysis tool using Classical Laminate Theory, Bredt-Batho torsion, and Tsai-Wu failure criteria to analyze the wing under load; design confirmed by a physical 5G wing loading test
- Selected composite materials, carbon-fiber tube interfaces, and bonding agents for a lightweight UAV wing, and designed PA-612 CF15 printed ribs, motor mounts, and fixtures by iterating print orientation, support strategy, warping control, fit, and dimensional accuracy for functional aircraft hardware
- Performed composite hand layup, vacuum bagging, bonding, and structural validation for the critical wing build

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Technical Skills

<u>CAD / Design:</u> SolidWorks, fixture design, bracket design, structural design

<u>Analysis / Testing:</u> Structural/composite analysis, system verification, hardware validation, durability testing, test fixture development

<u>Manufacturing / Prototyping:</u> FDM 3D printing, MSLA 3D printing, DfAM, composite hand layup, vacuum bagging, prototype assembly, hand fabrication

<u>Programming / Integration:</u> Python, Flask, REST API integration, LabJack I/O, Git

<u>Materials:</u> Nylon PA-612 CF15, composites, carbon-fiber tubes, carbon-fiber and fiberglass fabrics, 3D-printing polymers, adhesive bonding

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Leadership

Eagle Scout